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EVALUATION

I. Choose the best answer.

1. 40 ml of methane is completely burnt using 80 ml of oxygen at room temperature The volume of gas left after cooling to room temperature is

   (a) 40 ml CO_2 gas

   (b) 40 ml CO_2 gas and 80 ml H_2O gas

   (c) 60 ml CO_2 gas and 60 ml H_2O gas

   (d) 120 ml CO_2 gas

2. An element X has the following isotopic composition ^{200}X = 90 %, ^{199}X = 8 % and ^{202}X = 2 %. The weighted average atomic mass of the element X is closest to

   (a) 201 u

   (b) 202 u

   (c) 199 u

   (d) 200 u

3. Assertion : Two mole of glucose contains 12.044×10^{23} molecules of glucose

   Reason : Total number of entities present in one mole of any substance is equal to 6.02×10^{22}

   (a) both assertion and reason are true and the reason is the correct explanation of assertion

   (b) both assertion and reason are true but reason is not the correct explanation of assertion

   (c) assertion is true but reason is false

   (d) both assertion and reason are false

4. Carbon forms two oxides, namely carbon monoxide and carbon dioxide. The equivalent mass of which element remains constant?

   (a) Carbon

   (b) oxygen

   (c) both carbon and oxygen

   (d) neither carbon nor oxygen

5. The equivalent mass of a trivalent metal element is 9 g eq^{-1} the molar mass of its anhydrous oxide is

   (a) 102 g (b) 27 g (c) 270 g (d) 78 g